RAG Learning

2.1 Algebraic Expressions

OCR GCSE Mathematics (J560) — Paper 1, 2 & 3 — Foundation and Higher

2.1.1 Algebraic Notation and Writing Expressions

Key Vocabulary

Term: A single number, variable, or product of numbers and variables. E.g. $3x$, $-5y^2$, 7.

Coefficient: The numerical factor of a term. In $-4x^3$ the coefficient is -4.

Variable: A letter representing an unknown or changing quantity, e.g. x, y, t.

Constant: A term with no variable part; its value does not change, e.g. 6 in $3x + 6$.

Expression: A combination of terms without an equals sign, e.g. $5x^2 - 3x + 2$.

Equation: Two expressions connected by an equals sign, e.g. $2x + 1 = 9$.

Formula: A rule connecting two or more variables, e.g. $A = \pi r^2$.

Identity: True for all values of the variable, shown with $\equiv$, e.g. $2(x + 3) \equiv 2x + 6$.

Writing Algebraic Expressions from Words

Word phrase	Algebraic expression
n more than 5	$5 + n$

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3 less than x	$x - 3$
double y	$2y$
k squared	k^2
p divided by 4	$\frac{p}{4}$
product of a and b	ab
sum of m and n, all multiplied by 3	$3(m+n)$
square root of q	$\sqrt{q}$

Exam Tip: Never write multiplication signs between a number and a letter: $3 \times x$ is written $3x$. Also, $1 \times x = x$ (never write $1x$ unless expanding a bracket).

Understanding Index Notation

In algebra, powers follow the same rules as arithmetic:

$$a^m \times a^n = a^{m+n}, \qquad a^m \div a^n = a^{m-n}, \qquad (a^m)^n = a^{mn}$$

$$a^0 = 1 \quad (a \neq 0), \qquad a^{-n} = \frac{1}{a^n}, \qquad a^{1/n} = \sqrt[n]{a}$$

Worked Example: Simplify $\frac{x^5 \cdot x^{-2}}{x^3}$.

> Numerator: $x^5 \cdot x^{-2} = x^{5+(-2)} = x^3$
>
> Overall: $\frac{x^3}{x^3} = x^0 = 1$

2.1.2 Simplifying Expressions

Collecting Like Terms

Like terms have exactly the same variable part (including power). Only like terms can be added or subtracted.

- $3x$ and $7x$ are like terms → $3x + 7x = 10x$
- $5y^2$ and $-2y^2$ are like terms → $5y^2 - 2y^2 = 3y^2$
- $4x$ and $4x^2$ are *not* like terms (different powers)
- $3xy$ and $5yx$ are like terms ($xy = yx$)

Worked Example: Simplify $7a + 3b - 2a + 5b - b$.

> Group a terms: $7a - 2a = 5a$
>
> Group b terms: $3b + 5b - b = 7b$
>
> Answer: $5a + 7b$

Worked Example: Simplify $4x^2 - 3x + 2 + x^2 + 5x - 7$.

> x^2 terms: $4x^2 + x^2 = 5x^2$
>
> x terms: $-3x + 5x = 2x$
>
> Constants: $2 - 7 = -5$
>
> Answer: $5x^2 + 2x - 5$

Multiplying and Dividing Terms

When multiplying terms, multiply the coefficients and add the indices of like bases:

$$3x^2 \times 4x^3 = 12x^5 \qquad (\text{multiply coefficients: } 3 \times 4 = 12, \text{ add powers: } 2 + 3 = 5)$$

When dividing terms, divide the coefficients and subtract the indices:

$$\frac{15x^4}{5x} = 3x^3 \qquad (\text{divide coefficients: } 15 \div 5 = 3, \text{ subtract powers: } 4 - 1 = 3)$$

Worked Example: Simplify $\frac{6a^3b^2}{9a^2b^4}$.

> Coefficients: $6 \div 9 = \frac{2}{3}$
>
> a terms: $a^{3-2} = a$
>
> b terms: $b^{2-4} = b^{-2} = \frac{1}{b^2}$
>
> Answer: $\frac{2a}{3b^2}$

Exam Tip: Multiplying two negatives gives a positive: $(-3x)(-4y) = 12xy$. Take extra care with signs.

2.1.3 Expanding Brackets

Single Brackets

Multiply every term inside the bracket by the term outside:

$$a(b + c) = ab + ac$$

Worked Example: Expand $5(3x - 4)$.

> $5 \times 3x = 15x$
>
> $5 \times (-4) = -20$
>
> Answer: $15x - 20$

Worked Example: Expand and simplify $3(2x + 1) - 2(x - 5)$.

> $3(2x + 1) = 6x + 3$

> $-2(x-5) = -2x + 10$
>
> Combine: $6x + 3 - 2x + 10 = 4x + 13$

Double Brackets — FOIL Method

To expand $(a+b)(c+d)$, multiply each term in the first bracket by each term in the second:

$$\underbrace{ac}_{\text{F}} + \underbrace{ad}_{\text{O}} + \underbrace{bc}_{\text{I}} + \underbrace{bd}_{\text{L}}$$

> **Worked Example:** Expand $(x+3)(x+5)$.
>
> F: $x \cdot x = x^2$
>
> O: $x \cdot 5 = 5x$
>
> I: $3 \cdot x = 3x$
>
> L: $3 \cdot 5 = 15$
>
> Collect like terms: $x^2 + 5x + 3x + 15 = x^2 + 8x + 15$

Grid Method for Double Brackets

The grid method is a reliable alternative to FOIL. Each cell is the product of the row and column headers.

Grid method: (2x + 3)(x − 4)

	2x	+3
x	$2x^2$	3x
−4	−8x	−12

$= 2x^2$
$+ 3x$
$- 8x$
$- 12$

$= 2x^2 - 5x - 12$

Special Cases

Perfect square: $(a+b)^2$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

Worked Example: Expand $(3x-2)^2$.

> $(3x)^2 - 2(3x)(2) + 2^2$
>
> $= 9x^2 - 12x + 4$

Difference of two squares: $(a+b)(a-b)$

$$(a+b)(a-b) = a^2 - b^2$$

Worked Example: Expand $(5x+3)(5x-3)$.

> $(5x)^2 - 3^2 = 25x^2 - 9$

Exam Tip: Recognising the difference of two squares is a key skill — both in expanding and factorising. Look for two brackets that are identical except for the sign between the terms.

2.1.4 Factorising

Taking Out the HCF

Find the highest common factor (HCF) of all terms and place it outside a bracket:

Worked Example: Factorise $12x^3 - 8x^2 + 4x$.

> HCF of 12, 8, 4 is 4; HCF of x^3, x^2, x is x
>
> HCF = $4x$

> $4x(3x^2 - 2x + 1)$
>
> Check by expanding: $4x \cdot 3x^2 = 12x^3$ ✓

Worked Example: Factorise $6a^2b - 9ab^2$.

> HCF = $3ab$
>
> $3ab(2a - 3b)$

Factorising Quadratics $ax^2 + bx + c$

Case 1: $a = 1$ **(simple quadratics)**

Find two numbers that *multiply* to give c and *add* to give b:

Worked Example: Factorise $x^2 + 7x + 12$.

> Need two numbers that multiply to $+12$ and add to $+7$
>
> Factors of 12: $(1, 12), (2, 6), (3, 4) \rightarrow 3 + 4 = 7$ ✓
>
> Answer: $(x + 3)(x + 4)$

Worked Example: Factorise $x^2 - 5x - 6$.

> Need two numbers that multiply to -6 and add to -5
>
> -6 and $+1$: $(-6)(+1) = -6$ ✓, $-6 + 1 = -5$ ✓
>
> Answer: $(x - 6)(x + 1)$

Sign pattern in $x^2 + bx + c$	Signs in factors	Example
$b > 0, c > 0$	Both positive	$(x + 3)(x + 4)$

$b < 0, c > 0$	Both negative	$(x-3)(x-4)$
$c < 0$	One positive, one negative	$(x+3)(x-4)$ or $(x-3)(x+4)$

Case 2: $a \neq 1$ — The AC Method

1. Find the product ac.
2. Find two numbers that multiply to ac and add to b.
3. Rewrite the middle term using these two numbers, then factorise by grouping.

Worked Example: Factorise $2x^2 + 7x + 3$.

> $a = 2,\ b = 7,\ c = 3$ so $ac = 6$
>
> Find two numbers: multiply to 6, add to 7 → 1 and 6
>
> Rewrite: $2x^2 + x + 6x + 3$
>
> Group: $x(2x+1) + 3(2x+1)$
>
> Factor: $(2x+1)(x+3)$

Difference of Two Squares (Factorising)

$$a^2 - b^2 = (a+b)(a-b)$$

Worked Example: Factorise $9x^2 - 16$.

> $9x^2 = (3x)^2$ and $16 = 4^2$
>
> Answer: $(3x+4)(3x-4)$

Exam Tip: Always check for a common factor first before using any other factorising method. E.g. $2x^2 - 8 = 2(x^2-4) = 2(x+2)(x-2)$.

Higher — Factorising by Grouping (4-term expressions)

When an expression has four terms, try grouping them in pairs and factorising each pair:

Worked Example: Factorise $ax + ay + bx + by$.

- Group: $(ax + ay) + (bx + by)$
- Factorise each pair: $a(x + y) + b(x + y)$
- The common bracket is $(x + y)$: $(x + y)(a + b)$

Worked Example: Factorise $3x^2 - 6x + 2x - 4$.

- Group: $3x(x - 2) + 2(x - 2)$
- Answer: $(x - 2)(3x + 2)$

2.1.5 Algebraic Fractions [Higher]

Higher only

Simplifying Algebraic Fractions

Factorise the numerator and denominator completely, then cancel common factors:

Worked Example: Simplify $\dfrac{x^2 - 9}{x^2 - x - 6}$.

- Numerator: $x^2 - 9 = (x + 3)(x - 3)$
- Denominator: $x^2 - x - 6 = (x - 3)(x + 2)$

Cancel $(x-3)$: $\frac{(x+3)\cancel{(x-3)}}{\cancel{(x-3)}(x+2)} = \frac{x+3}{x+2}$

Multiplying and Dividing Algebraic Fractions

Multiply: multiply numerators together and denominators together (simplify first if possible).

Divide: flip the second fraction (reciprocal) and multiply.

Worked Example: Simplify $\frac{3x+6}{x^2} \times \frac{x}{x+2}$.

Factorise: $\frac{3(x+2)}{x^2} \times \frac{x}{x+2}$

Cancel $(x+2)$ and one x: $\frac{3 \cdot x}{x^2} = \frac{3}{x}$

Adding and Subtracting Algebraic Fractions

Find the lowest common denominator (LCD), rewrite each fraction, then combine numerators:

Worked Example: Simplify $\frac{3}{x+1} + \frac{2}{x-2}$.

LCD $= (x+1)(x-2)$

Rewrite: $\frac{3(x-2)}{(x+1)(x-2)} + \frac{2(x+1)}{(x+1)(x-2)}$

Combine: $\frac{3(x-2)+2(x+1)}{(x+1)(x-2)}$

Expand numerator: $3x - 6 + 2x + 2 = 5x - 4$

Answer: $\frac{5x-4}{(x+1)(x-2)}$

Exam Tip: In Higher exams, algebraic fractions questions often require factorising before

simplifying. Always factorise both top and bottom first.

Quick Reference Summary

Skill	Key rule / formula
Collecting like terms	Only add/subtract terms with identical variable parts
Multiplying terms	Multiply coefficients, add indices: $x^m \cdot x^n = x^{m+n}$
Expand single bracket	$a(b+c) = ab + ac$
Expand double brackets	FOIL or grid method; collect like terms
Perfect square	$(a \pm b)^2 = a^2 \pm 2ab + b^2$
Difference of two squares	$(a+b)(a-b) = a^2 - b^2$
Factorise: HCF	Identify and extract highest common factor
Factorise: simple quadratic	Two numbers that multiply to c and add to b
Factorise: $a \neq 1$	AC method — split middle term, group
Algebraic fractions [H]	Factorise fully, then cancel common factors

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